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On an extension of the Landau-Gonek formula

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ABSTRACT

We prove an extension of the Landau-Gonek formula. As an application, unconditionally, we recover an important consequence of Montgomery's work on the pair correlation that previously was known under the Riemann Hypothesis. We proceed to show that at least two-thirds of the zeros of the zeta function are simple under a milder assumption than the Riemann Hypothesis. We will also prove a corollary on the value-distribution of Dirichlet polynomials.

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1. Introduction

For $\mathbf{Re}(s) > 1$, the Riemann zeta function is defined as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

It has an analytic continuation to the whole complex plane except $s = 1$ and non-trivial zeros in $0 < \mathbf{Re}(s) < 1$, known as the critical strip. The location of the zeros of the zeta function is closely tied to the distribution of prime numbers. The closer the zeros are to the critical line ($\mathbf{Re}(s) = \frac{1}{2}$), the better the control is over the distribution of the primes.

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The Riemann Hypothesis (RH) predicts that all of the non-trivial zeros of the Riemann zeta function are located on this line.

One of the ways that we can see how the prime numbers are connected to the zeros of the zeta function is to look at them through the lens of the Landau-Gonek formula.

In 1911 Landau [8] proved that for fixed $x > 1$

$$\sum_{\substack{\zeta(\rho)=0 \\ 0<\mathrm{Im}(\rho)<T}} x^{\rho} = -\frac{T}{2\pi}\Lambda(x) + O(\log T),$$

where $\Lambda(x) = \log p$, if x equals to a power of prime p, and 0 otherwise. Later Gonek [12] proved a uniform version of Landau's formula: Let $x, T > 1$, and for simplicity consider x to be an integer. Then

$$\sum_{\substack{\zeta(\rho)=0 \\ 0<\mathrm{Im}(\rho)<T}} x^{\rho} = -\frac{T}{2\pi}\Lambda(x) + O\big(x \log 2xT \log\log 3x\big). \tag{1.1}$$

The formula shows that one can test whether or not x is a prime number by using an averaging sum involving the zeros of the zeta function. The formula has applications on the statistical distribution of the zeros of the zeta function. As an example, Gonek reproved the following result of Montgomery [9].

Theorem *(Montgomery). Assume the Riemann Hypothesis and fix* $0 < \alpha < 1$. *Then*

$$\sum_{0<\gamma,\gamma'<T} \left(\frac{\sin(\frac{\alpha}{2}(\gamma-\gamma')\log T)}{\frac{\alpha}{2}(\gamma-\gamma')\log T}\right)^2 \sim \big(\frac{1}{\alpha} + \frac{\alpha}{3}\big)\frac{T}{2\pi}\log T, \tag{1.2}$$

where γ *denotes the imaginary part of a zeta zero.*

In Corollary 1.2 we will prove the above unconditionally. Montgomery proved this as a part of his work on the pair correlation of the zeros of the Riemann zeta function. We should emphasize that interesting applications in his work emerged from (1.2).

In this paper, we extend the connections between the zeros of the Riemann zeta function and prime numbers. To explain the result we assume RH, although our result is independent of it. Consider a function $C_\alpha(\rho)$ that counts the number of zeros of the zeta function that are close (scaled by α) to ρ.

$$C_\alpha(\rho) = \#\{\rho' : |\rho - \rho'| \le \alpha/\log T\}.$$

Now imagine that we add this function to the Landau-Gonek formula, in a dot product manner $\langle C_\alpha(\rho), x^\rho\rangle$, to have

$$\sum_{\substack{\zeta(\rho)=0\\ 0<\mathrm{Im}(\rho)<T}} C_\alpha(\rho)x^\rho. \tag{1.3}$$

We show that for $\alpha \gg 1$ the above sum is basically the same as the Landau-Gonek formula, however changes start to emerge when α gets smaller. For small α, adding the factor C_α in the Landau-Gonek formula would extend its support from primes to products of two primes. Before stating our theorem, let us give formal definitions of objects we work with.

Definition. Let s, ρ be complex numbers. Define

$$\mathcal{D}_{\alpha,\rho}(s) := \left(\frac{e^{\alpha \log T \frac{(s-\rho)}{2}} - e^{-\alpha \log T \frac{(s-\rho)}{2}}}{s-\rho}\right)^2, \tag{1.4}$$

where if $s=\rho$, we define $\mathcal{D}_{\alpha,\rho}(s) = (\alpha \log T)^2$. Let

$$C_{\alpha,\rho} = \sum_{\zeta(\rho')=0} \mathcal{D}_{\alpha,\rho}(\rho'). \tag{1.5}$$

Also for $\Delta > 0$, define

$$\omega(s) := \frac{1}{\sqrt{\pi}\Delta}\, e^{\frac{(s-(1/2+iT))^2}{\Delta^2}}. \tag{1.6}$$

Throughout the paper we take $\Delta = T(\log T)^{-1}$.

Theorem 1.1. *For $\alpha < 1$, let x be of the form r/s where r, s are integers smaller than $T^{1-\alpha}\log^{-5} T$. For $x > 1$ we have that*

$$\sum_{\zeta(\rho)=0} \omega(\rho) C_{\alpha,\rho}\, x^\rho + O\left(\frac{1}{T^{1-\alpha}}\right)$$

$$= \begin{cases} \frac{\log p \log q}{2\pi}\left(\log\left(\frac{T^\alpha}{p^i}\right)\mathbb{1}_{p^i<T^\alpha} + \log\left(\frac{T^\alpha}{q^i}\right)\mathbb{1}_{q^i<T^\alpha}\right) & \text{if } x = p^i q^j, \\ -\frac{\log T \log p}{2\pi}\left(\alpha \log T - \log\left(\frac{T^\alpha}{p^i}\right)\mathbb{1}_{p^i<T^\alpha}\right) & \text{if } x = p^i, \\ \frac{\log p \log q}{2\pi q^j}\left(\log\left(\frac{T^\alpha}{p^i}\right)\mathbb{1}_{p^i<T^\alpha} + \log\left(\frac{T^\alpha}{q^i}\right)\mathbb{1}_{q^i<T^\alpha}\right) & \text{if } x = p^i q^{-j}, \\ 0 & \text{otherwise,} \end{cases}$$

where ω is an smoothing weight defined in (1.6) with $\Delta = T(\log T)^{-1}$. For the case $x = 1$, we have

$$\frac{1}{\log^3 T}\sum_{\zeta(\rho)=0}\omega(\rho)C_{\alpha,\rho}=\frac{1}{6\pi}\alpha^3+\frac{\alpha}{2\pi}+O(\tfrac{1}{\log^3 T}). \tag{1.7}$$

By setting $z=(s-\rho)/2i$, we get that $\mathcal{D}_{\alpha,\rho}(s)$ in (1.4) equals to

$$\Big(\frac{\sin(\alpha\log Tz)}{z}\Big)^2,$$

using the complex definition of the Sine function. This is also known as the Fejer kernel. Therefore we can write (1.7) as the following.

Corollary 1.2. *Let $0<\alpha<1$, and consider the Sine function as a complex valued function. For the weighted version we have*

$$\sum_{\zeta(\rho)=0}\omega(\rho)\sum_{\zeta(\rho')=0}\left(\frac{\sin(\frac{\alpha}{2i}(\rho-\rho')\log T)}{\frac{\alpha}{2i}(\rho-\rho')\log T}\right)^2=\frac{\log T}{2\pi}(\frac{1}{\alpha}+\frac{\alpha}{3})+O(\frac{1}{\log^2 T}).$$

For the not-weighted version we have: Let $\alpha<1-\log\log T(\log T)^{-1}$, then

$$\frac{2\pi}{T\log T}\sum_{0<\gamma,\gamma'<T}\left(\frac{\sin(\frac{\alpha}{2i}(\rho-\rho')\log T)}{\frac{\alpha}{2i}(\rho-\rho')\log T}\right)^2=\frac{1}{\alpha}+\frac{\alpha}{3}+O(\frac{1}{\log T}). \tag{1.8}$$

Equation (1.8) above is the unconditional version of (1.2). Rudnick and Sarnak proved an smooth version of Montgomery pair correlation result, commenting that if the test function defined appropriately for complex values, then they can drop the assumption of RH. However they had to appeal back to assume RH for proving (1.8). See Theorem 3.2 and the first paragraph of the page 285 in [13]. The importance of (1.8) is that it does captures, to a certain extent, the microscopic behaviour of zeta zeros. This is significant because the most interesting corollaries follow from it.

Remark. We can obtain a similar result with considering two different L-functions. For example the sum over ρ can run among the zeros of the Riemann zeta function and the sum over ρ' can be considered to run among the zeros of a Dirichlet L-function.

1.1. Simple zeros of the Riemann zeta function

An important application of Montgomery's theorem (1.2) was to the problem of simple zeros of the Riemann zeta function. The conjecture is that all of the zeros are simple. Montgomery showed that assuming RH at least two-thirds of the zeros are simple. There were several improvements to this result, the best known conditional lower bound is by Bui and Heath-Brown [4] that shows that at least 70% of the zeros are simple. Unconditionally, the best result is due to Bui, Conrey, and Young [6]. They showed that more than 41% of the zeros are simple.

We relax the assumption of RH and get Montgomery's result assuming there are not too many zeros that violate RH. We define

$$N(\sigma, T) := \#\{\rho : \zeta(\rho) = 0 \text{ and } \mathbf{Re}(\rho) > \sigma \text{ and } |\mathbf{Im}(\rho)| \leq T\}. \tag{1.9}$$

Corollary 1.3. *For $\sigma > 1/2$, assume the zero density hypothesis*

$$N(\sigma, T) \ll T^{2(1-\sigma)} \log^{-B} T,$$

with $B > 3$. Then at least two-third of the zeros are simple.

We could also get the same corollary by assuming that the zeros of the Riemann zeta function satisfy a non-clustering assumption.

The problem of small gap between zeros of the Riemann zeta function is a central problem in number theory due to its connection to the class number formula. Let γ, γ' denote consecutive ordinates of zeros of the zeta function.

$$\mu := \liminf_{\gamma > 0} \frac{(\gamma - \gamma') \log \gamma}{2\pi}.$$

Unconditionally, we do not have much information, except Selberg in [14] showed $\mu < 1$. Conditional to RH, many authors [10,7,5,11,15] worked on the problem, and the value of μ is down to 0.515396. Getting μ smaller than 0.5 would have significant application in the class number problem, however the method used to obtain the mentioned results is known to be inadequate to resolve this problem. In [2] the author and Ng, proved that $\mu < 0.49999$ under the assumption of certain conjectural bounds on the auto-correlation of the von Mangoldt and the Liouville functions. Here we prove a result in this direction under a zero density hypothesis, rather than RH.

Corollary 1.4. *For $\sigma > 1/2$, assume the same zero density hypothesis as Corollary 1.3. Then there is a constant $\mu < 1$ such that the normalized gap between the imaginary parts of the consecutive zeros of the Riemann zeta function is getting smaller than μ times the average gap, infinitely often.*

Remark. With a back of the envelope calculation, if we assume $\mu > 0.78$, then for any γ,

$$\sum_{0 < \gamma' < T} \left(\frac{\sin(\frac{1}{2}(\gamma - \gamma') \log T)}{\frac{1}{2}(\gamma - \gamma') \log T} \right)^2 < 1.33,$$

which is at odds with Corollary 1.2. This implies that $\mu < 0.78$.

In [3] we look more closely at the application of the Fejer Kernel in the problem of small gaps between the zeros of the Riemann zeta function.

1.2. Value-distribution of special Dirichlet polynomials

We consider Dirichlet polynomials

$$A_1(s)=\sum_{\substack{n<T^{1-\alpha-\epsilon}\\ \lambda(n)=1}}\frac{1}{n^s}\quad\text{and}\quad A_2(s)=\sum_{\substack{n<T^{1-\alpha-\epsilon}\\ \lambda(n)=-1}}\frac{1}{n^s},\tag{1.10}$$

where $\lambda(\cdot)$ is the Liouville's function. These polynomials carry special properties with respect to zeros of the zeta function that often used to study the problem of gaps between zeros of zeta. For example for small α under RH we can show that, on average, we have $A_1=-A_2$ in vicinity of zeros.

Having said that, A_1 and A_2, individually, do not appear to correlate with the zeta zeros. By this, we mean that the average of $|A_1|^2,|A_2|^2$ on the zeros is the same as the average of them on the interval (say) $[1,T]$ i.e.

$$\frac{1}{N(T)}\sum_{0<\gamma\leq T}|A_i(\tfrac{1}{2}+i\gamma)|^2\sim\frac{1}{T}\int_1^T|A_i(\tfrac{1}{2}+it)|^2dt,\tag{1.11}$$

for $i=1,2$ and where $N(T)$ is the total number of zeros with imaginary part between 0 and T.

As a corollary of the Theorem 1.1 we prove that even though on the surface, they do not correlate with zeros, on a deeper level, they show significant correlation with accumulation of zeros. This means that they attain large values at zeros that either have high multiplicity or high accumulation in their vicinity.

Corollary 1.5. *Assume the Riemann Hypothesis. For $i=1,2$ have that*

$$\begin{aligned}\frac{2\pi}{\log^4 T}\sum_{\zeta(\rho)=0}\omega(\rho)C_{\alpha,\rho}\,|A_i(\rho)|^2&=\frac{1-\alpha}{8}\big(\frac{\alpha^3}{3}+\alpha\big)\\&+\int_0^{1-\alpha}\int_0^{(1-\alpha)-u}(2\alpha-u\mathbb{1}_{u<\alpha}-v\mathbb{1}_{v<\alpha})(1-\alpha-u-v)dudv.\end{aligned}\tag{1.12}$$

The second term in double integral gives the bias that come from the accumulation factor $C_{\alpha,\rho}$ in the LHS of (1.12). This means that if we bluntly insert what we get from (1.11) into the LHS of (1.12) we would only get the first term in the RHS of (1.12). For example, for $\alpha=1/3$, from the corollary we get

$$\frac{2\pi}{\log^4 T}\sum_{\zeta(\rho)=0}\omega(\rho)C_{\frac{1}{3},\rho}\,|A_i(\rho)|^2=0.05069\cdots,$$

which shows a bias of the amount about 0.021888 toward the accumulation of zeros. This result can be viewed as a part of our investigation of value-distribution of Dirichlet polynomials on the $\frac{1}{2}$-line which we started in [3].

2. Necessary Lemmas and the proof of main formulas

We begin this section by proving a smooth variant of Landau's formula.

Lemma 2.1. *Let $x > 0$ and $c > 1$, with ω defined in (1.6). Then*

$$\sum_{\zeta(\rho)=0} \omega(\rho) x^{\rho} = -\frac{x^{\frac{1}{2}+iT}}{2\pi} \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\frac{1}{2}+iT}} e^{-\Delta^2 \log^2(\frac{x}{n})} \tag{2.1}$$

$$- \frac{x^{\frac{1}{2}+iT}}{2\pi} \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\frac{1}{2}-iT}} e^{-\Delta^2 \log^2(xn)} - \int\limits_{(1-c)} \frac{\omega(s)}{2\pi i} x^s \frac{\chi'}{\chi}(s) ds.$$

For an integer m with $m \ll T(\log T)^{-2}$ the above reduces to

$$\sum_{\zeta(\rho)=0} \frac{\omega(\rho)}{m^{\rho}} = -\frac{\Lambda(m)}{2\pi m} + O(e^{-(\log T)^{1+\epsilon}}). \tag{2.2}$$

Before giving a proof of Lemma 2.1 we state the following lemma from the author's thesis [1, Lemma 3.6, page 59].

Lemma 2.2. *Let $c \in \mathbb{R}$ and $x > 0$. Then*

$$\frac{1}{2\pi i} \int\limits_{(c)} \omega(s) x^s ds = \frac{1}{2\pi} x^{\frac{1}{2}+iT} e^{-\frac{\Delta^2 \log^2 x}{4}}, \tag{2.3}$$

$$\frac{1}{2\pi i} \int\limits_{(c)} \omega(1-s) x^s ds = \frac{1}{2\pi} x^{\frac{1}{2}-iT} e^{-\frac{\Delta^2 \log^2 x}{4}}, \tag{2.4}$$

$$\int\limits_{-\infty}^{\infty} \omega(\tfrac{1}{2}+it) dt = 1. \tag{2.5}$$

Proof of Lemma 2.1. By the residue theorem for $c > 1$ we have

$$\sum_{\zeta(\rho)=0} \omega(\rho) x^{\rho} = \frac{1}{2\pi i} \int\limits_{c-i\infty}^{c+i\infty} \omega(s) x^s \frac{\zeta'}{\zeta}(s) ds - \frac{1}{2\pi i} \int\limits_{1-c-i\infty}^{1-c+i\infty} \omega(s) x^s \frac{\zeta'}{\zeta}(s) ds$$

By using the logarithmic derivative of the functional equation for zeta

$$\frac{\zeta'}{\zeta}(s) = \frac{\chi'}{\chi}(s) - \frac{\zeta'}{\zeta}(1-s). \tag{2.6}$$

we can take the integral on the line $\mathbf{Re}(s) = 1-c$ back to the line $\mathbf{Re}(s) = c$ and since $c > 1$ we can use the series expansion of $\frac{\zeta'}{\zeta}$. Using Lemma 2.2 we have

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \omega(s)x^s\frac{\zeta'}{\zeta}(s)ds = -\frac{x^{\frac{1}{2}+iT}}{2\pi}\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^{\frac{1}{2}+iT}}e^{-\Delta^2\log^2(\frac{x}{n})}.$$

For the integral on the line $\mathbf{Re}(s) = 1-c$ we have

$$\frac{1}{2\pi i}\int_{(1-c)} \omega(s)x^s\frac{\zeta'}{\zeta}(s)ds = -\frac{x^{\frac{1}{2}+iT}}{2\pi}\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^{\frac{1}{2}-iT}}e^{-\Delta^2\log^2(xn)} - \int_{(1-c)}\frac{\omega(s)}{2\pi i}x^s\frac{\chi'}{\chi}(s)ds.$$

This completes the proof of the lemma. □

We need the following lemma in the proof of our theorem:

Lemma 2.3. *Let $x = r/s$, with $1 < r, s < T/\log^2 T$ and $x > 1$, we have*

$$\sum_{\zeta(\rho)=0}\omega(\rho)x^\rho\log(t_\rho/2\pi) = -\frac{\Lambda(x)}{2\pi}\int\omega(\tfrac{1}{2}+it)\frac{\chi'}{\chi}(\tfrac{1}{2}+it)dt + O(\frac{x^{1+\epsilon}}{T^2}). \tag{2.7}$$

Proof. For $c > 1$

$$\sum_{\zeta(\rho)=0}\omega(\rho)x^\rho\frac{\chi'}{\chi}(\rho) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\omega(s)x^s\frac{\zeta'}{\zeta}(s)\frac{\chi'}{\chi}(s)ds - \frac{1}{2\pi i}\int_{1-c-i\infty}^{1-c+i\infty}\omega(s)x^s\frac{\zeta'}{\zeta}(s)\frac{\chi'}{\chi}(s)ds. \tag{2.8}$$

Since $c > 1$ we have

$$\int_{c-i\infty}^{c+i\infty}\omega(s)x^s\frac{\zeta'}{\zeta}(s)\frac{\chi'}{\chi}(s)ds = -\sum_{n=1}^{\infty}\Lambda(n)\int_{c-i\infty}^{c+i\infty}\omega(s)(x/n)^s\frac{\chi'}{\chi}(s).$$

If $n = x$ that is

$$\Lambda(n)\int\omega(\tfrac{1}{2}+it)\frac{\chi'}{\chi}(\tfrac{1}{2}+it)dt.$$

For $x \neq n$ we will use the following expansion for the gamma factor

$$\frac{\chi'}{\chi}(\sigma_\rho + it) = -\log\big(\frac{t}{2\pi}\big) - \frac{i(1-2\sigma_\rho)}{t} + O\big(\frac{1}{t^2}\big) \tag{2.9}$$

and therefore we need to estimate

$$\int_{c-i\infty}^{c+i\infty} \omega(c+it)(x/n)^{c+it}\Big(\log(t/2\pi) + O(t^{-2})\Big)dt.$$

By using integration by parts and the fact that $x = r/s < T/\log^2 T$, we can show that

$$\int_{c-i\infty}^{c+i\infty} \omega(c+it)(r/sn)^{c+it}\log(t/2\pi)\ dt,$$

is very small. By using trivial bound on the error term we have

$$\int_{c-i\infty}^{c+i\infty} \omega(c+it)(x/n)^{c+it}O\big(\frac{1}{t^2}\big)\ dt \ll \frac{x^c}{T^2}.$$

For the integral on the line $\mathbf{Re}(s) = 1-c$ we use (2.6) and we get

$$\int_{1-c-i\infty}^{1-c+i\infty} \omega(s)x^s\big(\frac{\chi'}{\chi}(s) - \frac{\zeta'}{\zeta}(1-s)\big)\frac{\chi'}{\chi}(s)ds.$$

In this case we do not have a diagonal term and its bounded by $O(T^{-2}x^{1-c})$. □

We prove a lemma similar to Lemma 2.1 but instead of a Gaussian weight we use a generalization of the Fejer kernel defined in (1.4).

Lemma 2.4. *Let $\alpha > 0$ and $\mathcal{D}_{\alpha,\rho}(s)$ be as (1.4). For $y > 0$ and $\rho = \sigma + i\gamma$, we have*

$$\frac{1}{2\pi i}\int_{(c)} y^s\mathcal{D}_{\alpha,\rho}(s)ds = \begin{cases} y^\rho\log(T^\alpha y) & \text{if } \ 1 > y > T^{-\alpha}, \\ y^\rho\log(T^\alpha/y) & \text{if } \ T^\alpha > y \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

We also have

$$\frac{1}{2\pi i}\int_{(c)} y^s\mathcal{D}_{\alpha,\rho}(1-s)ds = \begin{cases} y^{1-\rho}\log(T^\alpha y) & \text{if } \ 1 > y > T^{-\alpha}, \\ y^{1-\rho}\log(T^\alpha/y) & \text{if } \ T^\alpha > y \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. For simplicity let us take $Q = T^{\alpha}$. We have that

$$\mathcal{D}_{\alpha,\rho}(s) = \frac{Q^{s-\rho}}{(s-\rho)^2} + \frac{Q^{\rho-s}}{(s-\rho)^2} - \frac{2}{(s-\rho)^2}.$$

Therefore for $c \neq \sigma$ we set $s - \rho = z$ and we have

$$\int y^s \mathcal{D}_{\alpha,\rho}(s) = \int y^{\rho} \frac{(Qy)^z}{z^2} + y^{\rho} \frac{(y/Q)^z}{z^2} - y^{\rho} \frac{2y^z}{z^2}.$$

To continue we use the following version of the Perron's formula

$$\frac{1}{2\pi i} \int\limits_{(c)} x^z \frac{ds}{z^2} = \begin{cases} \log x & \text{if} \quad x > 1, \\ 0 & \text{otherwise.} \end{cases}$$

For $1 > y > 1/Q$ we get $\log Qy$, for $Q > y > 1$ we get $\log Qy - 2\log y$, and for $y > Q$ everything will cancel out. This finishes the proof of the first part. For the second part note that

$$\int y^s \mathcal{D}_{\alpha,\rho}(1-s)ds = \int y^s \Big(\frac{Q^{\frac{s-(1-\rho)}{2}} - Q^{-\frac{s-(1-\rho)}{2}}}{s-(1-\rho)} \Big)^2 ds,$$

therefore the proof is the same just we need to switch ρ to $1-\rho$. □

We apply the above lemma to get

Lemma 2.5. *For $x > 0$, we have*

$$\begin{aligned} \sum_{\zeta(\rho')=0} \mathcal{D}_{\alpha,\rho}(\rho') x^{\rho'} = &-x^{\rho} \sum_{x \leq n < T^{\alpha} x} \frac{\Lambda(n)}{n^{\rho}} \log\big(\frac{T^{\alpha} x}{n}\big) - x^{\rho} \sum_{n < T^{\alpha}/x} \frac{\Lambda(n)}{n^{1-\rho}} \log\big(\frac{T^{\alpha}}{nx}\big) \\ &- x^{\rho} \sum_{x/T^{\alpha} < n < x} \frac{\Lambda(n)}{n^{\rho}} \log\big(T^{\alpha} \frac{n}{x}\big) - \int\limits_{(1-c)} x^s \frac{\chi'}{\chi}(s) \mathcal{D}_{\alpha,\rho}(s) \frac{ds}{2\pi i} \\ &+ \mathcal{D}_{\alpha,\rho}(1) x. \end{aligned} \tag{2.10}$$

For $x = 1$ this simplifies to

$$\begin{aligned} \sum_{\zeta(\rho')=0} \mathcal{D}_{\alpha,\rho}(\rho') = &- \sum_{n < T^{\alpha}} \frac{\Lambda(n)}{n^{\rho}} \log\big(\frac{T^{\alpha}}{n}\big) - \sum_{n < T^{\alpha}} \frac{\Lambda(n)}{n^{1-\rho}} \log\big(\frac{T^{\alpha}}{n}\big) \\ &- \int\limits_{(1-c)} \frac{\chi'}{\chi}(s) \mathcal{D}_{\alpha,\rho}(s) \frac{ds}{2\pi i} + \mathcal{D}_{\alpha,\rho}(1). \end{aligned} \tag{2.11}$$

Proof. The proof goes similarly to the proof of Lemma 2.1. By the residue theorem, for $c > 1$ we have

$$\sum_{\zeta(\rho')=0} \mathcal{D}_{\alpha,\rho}(\rho')x^{\rho'} = \frac{1}{2\pi i}\int_{(c)} \mathcal{D}_{\alpha,\rho}(s)x^s\frac{\zeta'}{\zeta}(s)ds - \frac{1}{2\pi i}\int_{(1-c)} \mathcal{D}_{\alpha,\rho}(s)x^s\frac{\zeta'}{\zeta}(s)ds.$$

For the integral on $\mathbf{Re}(s) = c$ we expand the logarithmic derivative of zeta and apply Lemma 2.4 to get the result. For the integral on $\mathbf{Re}(s) = 1 - c$ we use the functional equation (2.6) and then apply Lemma 2.4. □

Next we give proof of the theorem and Corollary 1.2. An important point in how we avoid assuming the Riemann hypothesis in the proof is that, if we have a zero off the critical line, we center our weight exactly at that zero, rather than centering it at the point on the critical line with the same imaginary part. This allows us to transform the sum over primes back to the means of Landau-Gonek formula.

Proof of Theorem 1.1. We begin with multiplying (2.11) in Lemma 2.5 with $\omega(\rho)x^\rho$ and sum over ρ. Therefore we get

$$\begin{aligned}\sum_{\zeta(\rho)=0}\omega(\rho)x^\rho\sum_{\zeta(\rho')=0}\mathcal{D}_{\alpha,\rho}(\rho') &= \sum_{n<T^\alpha}\Lambda(n)\log\big(\frac{T^\alpha}{n}\big)\Big(-\sum\omega(\rho)(\tfrac{x}{n})^\rho\Big)\\ &+\sum\mathcal{D}_{\rho,T}(1)\omega(\rho)x^\rho + \sum_{n<T^\alpha}\frac{\Lambda(n)}{n}\log\big(\frac{T^\alpha}{n}\big)\Big(-\sum\omega(\rho)(xn)^\rho\Big)\\ &+\sum\omega(\rho)x^\rho\int_{(1-c)}\frac{\chi'}{\chi}(s)\mathcal{D}_{\alpha,\rho}(s)\frac{ds}{2\pi i}\end{aligned}$$

If x equals a prime smaller than T^α, ($x/n = 1$ in the first sum) then we get $-(2\pi)^{-1}\Lambda(x)\log T\log(T^\alpha/x)$.

Now assume that $x/n \neq 1$ we use the Lemma 2.1 on sums over ρ, and given $x = r/s$ we have the first term above equal to

$$\frac{1}{2\pi}\sum_{n<T^\alpha}\Lambda(n)\log\big(\frac{T^\alpha}{n}\big)\big(\frac{r}{sn}\big)^{\frac12+iT}\sum_m\frac{\Lambda(m)}{m^{\frac12+iT}}e^{-\Delta^2\log^2(\frac{r}{smn})} + \frac{\Lambda(m)}{m^{\frac12-iT}}e^{-\Delta^2\log^2(\frac{rm}{sn})}$$

plus a negligible error term coming from the gamma factor in Lemma 2.1. Recall that we took $\Delta = T(\log T)^{-1}$. We are going to show that the contribution from the sum over m, n would be just diagonal, coming from $r = smn$ and $rm = sn$. To show this, by the assumption we have $sn < T^\alpha r$, and since we assumed $r < T^{1-\alpha}\log^{-5}T$ we get $sn < T\log^{-5}T$, and because of this the off-diagonal contribution is negligible. By the above argument we get

$$\frac{\Lambda(m)\Lambda(n)}{2\pi}\big(\log(\frac{T^\alpha}{m})\mathbb{1}_{m<T^\alpha} + \log(\frac{T^\alpha}{n})\mathbb{1}_{n<T^\alpha}\big),$$

for $r/s = mn$ and we get

$$\frac{\Lambda(m)\Lambda(n)}{2\pi m}\log(\frac{T^\alpha}{n})\mathbb{1}_{n<T^\alpha},$$

for $r/s = n/m$. The other terms can be handled similarly. Now we will take care of the Gamma factor:

$$\sum_\rho \omega(\rho)(\frac{r}{s})^\rho \int\limits_{(1-c)} \frac{\chi'}{\chi}(z)\mathcal{D}_{\alpha,\rho}(z)\frac{dz}{2\pi i}.$$

We take the integral to $\mathbf{Re}(z) = \mathbf{Re}(\rho)$ and we get an error term of size $T^{\alpha\mathbf{Re}(\rho)}|\rho|^{-2}$. Considering the support of $\omega(\rho)$ this is bounded by $T^{-2+\alpha}$. Next, we need to calculate

$$\int\limits_{(0)} \frac{\chi'}{\chi}(\rho+z)\mathcal{D}_{\alpha,0}(z)\frac{dz}{2\pi i}.$$

We will use the asymptotic expansion (2.9) and we get

$$\alpha\log T\sum\omega(\rho)(\tfrac{r}{s})^\rho\log(\tfrac{\gamma_\rho}{2\pi}) + O(\frac{\sum\omega(\rho)x^{\mathbf{Re}(\rho)}\log^5 T}{T}).$$

We apply the Lemma 2.3 and we have the contribution of the gamma factor is

$$-\alpha\log^2 T\frac{\Lambda(r/s)}{2\pi} + O(\frac{\sum\omega(\rho)x^{\mathbf{Re}(\rho)}\log^5 T}{T}).$$

The error is bounded by $T^{1-\alpha}$, this finishes the proof for $x \neq 1$. For $x = 1$ we first give the proof of weighted version and then explain the proof of (1.8). By using (2.11) we have

$$\begin{aligned}\sum_\rho\omega(\rho)\sum_{\zeta(\rho')=0}\mathcal{D}_{\alpha,\rho}(\rho') &= -\sum_{n<T^\alpha}\Lambda(n)\log\big(\frac{T^\alpha}{n}\big)\Big(\sum_\rho\frac{\omega(\rho)}{n^\rho}+\frac{\omega(\rho)}{n^{1-\rho}}\Big)\\ &\quad -\sum_\rho\omega(\rho)\int\limits_{(1-c)}\frac{\chi'}{\chi}(s)\mathcal{D}_{\alpha,\rho}(s)\frac{ds}{2\pi i}+\sum_\rho\omega(\rho)\mathcal{D}_{\alpha,\rho}(1).\end{aligned}$$

Now applying the smooth version of the Landau-Gonek formula and the same treatment of the gamma factor as the case for $x \neq 1$ we get that the expression above is equal to

$$\begin{aligned}&\frac{1}{\pi}\sum_{n<T^\alpha}\frac{\Lambda^2(n)}{n}\log\big(\frac{T^\alpha}{n}\big)+\frac{\alpha\log^3 T}{2\pi}+O(1)\\ &=\frac{1}{6\pi}(\alpha\log T)^3+\frac{\alpha}{2\pi}\log^3 T+O(1).\end{aligned}$$

For the not weighted version (1.8) we have

$$\sum_{0<\mathbf{Im}(\rho)<T}\sum_{\zeta(\rho')=0}\mathcal{D}_{\alpha,\rho}(\rho')=-\sum_{n<T^{\alpha}}\Lambda(n)\log\big(\frac{T^{\alpha}}{n}\big)\Big(\sum_{0<\mathbf{Im}(\rho)<T}\frac{1}{n^{\rho}}+\frac{1}{n^{1-\rho}}\Big)$$
$$-\sum_{0<\mathbf{Im}(\rho)<T}\int_{(1-c)}\frac{\chi'}{\chi}(s)\mathcal{D}_{\alpha,\rho}(s)\frac{ds}{2\pi i}+\sum_{0<\mathbf{Im}(\rho)<T}\mathcal{D}_{\alpha,\rho}(1). \tag{2.12}$$

Now we use the Gonek's formula [12] for $x\in\mathbb{Z}$

$$\sum_{0<\mathbf{Im}(\rho)<T}\frac{1}{x^{\rho}}=-\frac{T}{2\pi}\Big(\frac{\Lambda(x)}{x}\Big)+O(\log 2Tx\log\log 3x). \tag{2.13}$$

Applying this to (2.12) we have it equals

$$\frac{T}{\pi}\sum_{n<T^{\alpha}}\frac{\Lambda^2(n)}{n}\log\big(\frac{T^{\alpha}}{n}\big)-\sum_{0<\mathbf{Im}(\rho)<T}\Big(\mathcal{D}_{\alpha,\rho}(1)+\int_{(1-c)}\frac{\chi'}{\chi}(s)\mathcal{D}_{\alpha,\rho}(s)\frac{ds}{2\pi i}\Big),$$

plus and error term of size $T^{\alpha}\log^2 T$. For the sum over $\mathcal{D}_{\alpha,\rho}(1)$ it is easy to show that it is bounded by

$$\sum_{0<\mathbf{Im}(\rho)<T}T^{\alpha\mathbf{Re}(\rho)}|\rho|^{-2}\ll T^{\alpha}\log^2 T.$$

For the integral involving the Gamma factor we take the integral to $\mathbf{Re}(s)=\mathbf{Re}(\rho)$ and we get an error term of order $T^{\alpha}\log^2 T$. Therefore what left to consider is

$$\sum_{0<\mathbf{Im}(\rho)<T}\int\frac{\chi'}{\chi}(\mathbf{Re}(\rho)+it)\Big(\frac{\sin\big(\alpha\log T(t-\gamma_{\rho})\big)}{(t-\gamma_{\rho})}\Big)^2\frac{dt}{2\pi}. \tag{2.14}$$

Using the expansion (2.9) we have the above equals

$$\alpha\log T\sum_{0<\mathbf{Im}(\rho)<T}\log\gamma_{\rho}+O(\frac{1}{\gamma_{\rho}}). \tag{2.15}$$

A minor calculation gives that (2.15) equals

$$\alpha\log^2 T\frac{T\log T}{2\pi}+O(T\log T).$$

For the sum over primes we have, using the prime number theorem (PNT)

$$\sum_{n<T^{\alpha}}\frac{\Lambda^2(n)}{n}\log\big(\frac{T^{\alpha}}{n}\big)=(\alpha\log T)^3+O(1).$$

This finishes the proof. □

Next, we will give a proof of Corollaries 1.3 and 1.4. We will prove these corollaries by showing that the effect of the zeros that violate the Riemann hypothesis are negligible in the LHS (1.7).

Proof. We assume a zero density hypothesis in the following form

$$N(\sigma, T) \ll T^{2(1-\sigma)} \log^{-B} T. \tag{2.16}$$

Let us partition the set of zeros that off the critical line to

$$\rho_i = \{\rho, 1-\rho : \tfrac{1}{2} + \frac{i}{\log T} < \mathbf{Re}(\rho) \leq \tfrac{1}{2} + \frac{i+1}{\log T}\}$$
$$\rho_{i,m} = \{\rho \in \rho_i : m-1 < \mathbf{Im}(\rho) \leq m\}$$

Let $\rho = 1/2 + \sigma + i\gamma$ and $\rho' = 1/2 - \sigma' + i\gamma'$ then we have

$$\mathcal{D}_{\alpha,\rho}(\rho') = \frac{T^{\sigma+\sigma'+i(\gamma-\gamma')} + T^{-(\sigma+\sigma'+i(\gamma-\gamma'))} - 2}{(\sigma+\sigma'+i(\gamma-\gamma'))^2}.$$

Therefore if we consider the effect of zeros in $\rho_{i,m}$ and $\rho_{j,m'}$ together we get that its bounded by

$$\frac{T^{i+j/\log T}}{1+(m-m')^2} \log^2 T.$$

For $m - m' = h$, where m, m' are in the effective range of ω, we have that the total effect is bounded by

$$\log^2 T \sum_{i,j} \frac{T^{i+j/\log T}}{h^2} \sum_{m-m'=h} |\rho_{i,m}||\rho_{j,m'}|$$
$$\ll \sum_{i,j} \frac{T^{i+j/\log T}}{h^2} \Big(\sum_m |\rho_{i,m}|^2\Big)^{\frac{1}{2}} \Big(\sum_m |\rho_{j,m}|^2\Big)^{\frac{1}{2}} \tag{2.17}$$

Using the zero counting formula we have $|\rho_{i,m}| < \log T$, therefore

$$\sum_m |\rho_{i,m}|^2 \ll \log T \sum_m |\rho_{i,m}| < T^{1-\frac{2i}{\log T}} \log^{-B+1} T.$$

The last inequality is obtained by using the zero density hypothesis (2.16). Putting this back into (2.17) and apply it to

$$\sum_{\substack{\zeta(\rho)=0\\ \mathbf{Re}(\rho)\neq 1/2}} \omega(\rho) \sum_{\substack{\zeta(\rho')=0\\ \mathbf{Re}(\rho')\neq 1/2}} \mathcal{D}_{\alpha,\rho}(\rho') \tag{2.18}$$

we get that the total effect of zeros off the critical in the above line is bounded by

$$T \log^{-B+6} T$$

considering $|\omega(\rho)| \ll T^{-1} \log T$. Therefore for $B > 3$ this gets absorbed in the error term considering the main term in (1.7) has size $\log^3 T$. The rest of the proof goes the same as the proof of Corollary 2 in [9]. □

Proof of Corollary 1.5. We give the proof for A_1, since the proof for A_2 is similar. We have that

$$\sum_{\zeta(\rho)=0} \omega(\rho) C_{\alpha,\rho} \, |A_1(\rho)|^2 = \sum_{\zeta(\rho)=0} \omega(\rho) C_{\alpha,\rho} \sum_{\substack{m,n\\ \lambda(m)=\lambda(n)=1}} \frac{1}{m} \Big(\frac{m}{n}\Big)^{\rho} =$$

$$\sum_{\substack{m,n\\ \lambda(m)=\lambda(n)=1}} \frac{1}{m} \sum_{\zeta(\rho)=0} \omega(\rho) C_{\alpha,\rho} \Big(\frac{m}{n}\Big)^{\rho}. \tag{2.19}$$

For $m = n$ using Corollary 1.2 we get the first term in RHS of (1.12). For $m \neq n$ using Theorem 1.1 we get

$$\frac{1}{\pi} \sum_{\substack{pqn<T^{1-\alpha} \log^{-5} T\\ \lambda(n)=1}} \frac{\log p \log q}{npq} \Big(2\alpha \log T - \log p \mathbb{1}_{p<T^{\alpha}} - \log q \mathbb{1}_{q<T^{\alpha}}\Big). \tag{2.20}$$

Using PNT, we get the second term in the RHS in (1.12). □

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